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SEAT No. :

PE-561

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S.E. (Artificial Intelligence and Machine Learning) (Insem.)

COMPUTER NETWORKS

(2019 Pattern) (Semester - III) (218543)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Figures to the right indicate full marks.***
3) ***Neat diagrams must be drawn wherever necessary.***
4) ***Assume suitable data if necessary.***

Q1) a) Define Signal. Discuss Any two Types of signal. With diagram. **[5]**

b) What is Multiplexing? Differentiate between FDM and WDM. **[5]**

c) Prove Shannon's Hartley theorem with respect to channel capacity. **[5]**

OR

Q2) a) Define Noise. Discuss different types of noise. **[5]**

b) Describe Signal to Noise(S/N) Ratio along with its significance. **[5]**

c) Describe Electromagnetic Spectrum and its Application. **[5]**

Q3) a) Draw a TCP/IP Model and explain the transport Layer. **[5]**

b) Define addressing and explain logical and port addressing with an example. **[5]**

c) Discuss Peer to Peer Network architecture with a diagram. **[5]**

OR

Q4) a) Demonstrate following networks; **[6]**

i) LAN

ii) MAN

iii) WAN.

b) Explain the functions of the following layer of the ISO/OSI model with a diagram **[5]**

i) Network Layer

ii) Presentation Layer

c) Write a Short note on **[4]**

i) Router

ii) Switch

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